

Decision summary

REJECT

SIMULATED REJECT — no generic specification-adjusted gap

The simulated internal grade matrix shows equivalent qualified supply above target demand; generic capacity support would be non-additional.

SAU-H0-390210 · No intervention · Mode: Ministry simulation

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Polypropylene, in primary forms

HS H0 / 390210

Unchanged public conclusion

REJECT — generic capacity support

Target-specification demand 56 kt	Effective qualified capacity 104.49 kt
Published capability distance D* 0 ratio	Minimum effective support S* 0 SAR m

Decision conditions

Reopen only for a named specialty-grade exception supported by buyer evidence.

Kill conditions

Any proposed generic-capacity support is stopped while equivalent qualified capacity is available.

Evidence passport ledger: [P-WITS-390210](#) [P-SABIC](#)

Snapshot: PUBLIC-SAU-H0-390210-2026-08-31 · As of: 2026-08-31

Supporting evidence pack

1. [Identity and specification boundary](#)
2. [Demand and public trade signals](#)
3. [Domestic supply and qualified capacity](#)
4. [Gap diagnosis and false-positive controls](#)
5. [Capability and route alternatives](#)
6. [Economics and intervention](#)
7. [Competition and public policy](#)
8. [Complete rule execution ledger](#)
9. [Evidence passports and contradictions](#)
10. [Conditions, kill conditions and next facts](#)
11. [Authority and reproducibility](#)

Identity and specification boundary

Class D simulation · SYN-MINISTRY-PP-001

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Unchanged public conclusion

REJECT — generic capacity support

No intervention for generic capacity

Reject generic polypropylene capacity support; investigate only explicitly defined grade/application exceptions.

Decision conditions

- Only a named specialty grade/application exception may re-enter INVESTIGATE.

Kill conditions

- Imported and domestic products are shown equivalent and qualified
- No specification-adjusted gap exists

Next evidence actions

- Saudi tariff-line and invoice description
- Importer/offtaker application and qualification requirements
- Producer grade, application-equivalence and local-availability matrix
- Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

- Delivered economics for the named specification/application exception

Product identity

Source field	Value	Unit
HS revision	H0	
HS code	390210	
National tariff line	Unavailable Not established by the supplied evidence.	
Commercial name, English source	Polypropylene, in primary forms	
Commercial name, Arabic	بولي بروبيلين بأشكاله الأولية	
Application boundary	Generic capacity case; narrow grade/application exceptions remain unresolved.	
Target specification	Generic injection-grade polypropylene SYN-MINISTRY-PP-001::target_specification	
Standard	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	
Alloy	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	
Temper	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	
Application	general rigid packaging SYN-MINISTRY-PP-001::target_specification	
Zinc coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	g/m ²
Tin coating mass	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	g/m ²
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	mm

Source field	Value	Unit
Thickness	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	µm
Width	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::target_specification	mm
Customer qualification required	No SYN-MINISTRY-PP-001::target_specification	

Classification revision and concordance

Unavailable

Specification source spans

Unavailable

Demand and public trade signals

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SIMULATED — NOT MINISTRY EVIDENCE

Demand conclusion

Gross imports are a screening signal, not demand at the target specification.

Source field	Value	Unit
Base demand	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::demand	kt
Target-specification demand	56 SYN-MINISTRY-PP-001::demand	kt
Downside demand	51 SYN-MINISTRY-PP-001::demand	kt
Committed demand	0 SYN-MINISTRY-PP-001::demand	kt

Source field	Value	Unit
Announced demand	0 SYN-MINISTRY-PP-001::demand	kt
Commitment probability	Unavailable Not established by the supplied evidence. SYN-MINISTRY-PP-001::demand	ratio
Observation year	2024	year
Gross import value	92.3	USD m
Gross import quantity	56	kt
Named buyers	Unavailable Not established by the supplied evidence.	
Buyer concentration	Unavailable Not established by the supplied evidence.	
Demand timing	Unavailable Not established by the supplied evidence.	
Year	Import value, USD m	Import quantity, kt
2021	91.2	50.5
2023	91	67
2024	92.3	56

Trade coverage and measurement

Trade-flow basis	gross
Re-exports separated	No
Domestic-origin exports separated	No
Missing years	• 2022
Monthly partner/tariff-line detail available	No
Quantity is comparable	Yes
Execution ceiling	DEGRADED where continuity or grade resolution is required

Production, retained flows and re-exports

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SIMULATED — NOT MINISTRY EVIDENCE

Reference year	2024
Domestic production (kt)	Unavailable
Retained imports (kt)	Unavailable
Domestic-origin exports (kt)	Unavailable
Re-exports (kt)	Unavailable
Source evidence references	• None recorded; no completed review is implied.

Supplier diagnostics

Domestic supply and qualified capacity

Supply conclusion

Public producer observations; nameplate is not qualified capacity.

Named producer	Value
SABIC	Process family polypropylene
	Published process route Unavailable
	Published standards Unavailable
	Published product portfolio homo, random and impact products and compounds
	Observed nameplate capacity, t/y Unavailable
	Nominal-capacity status unresolved
	Nominal-capacity evidence reference Unavailable
	Evidence class C
P-SABIC	

Named producer	Value
Advanced Petrochemical	Process family polypropylene
	Published process route Unavailable
	Published standards Unavailable
	Public source note Jubail complex exceeded nameplate in Q1 2026
	Observed nameplate capacity, t/y 450000
	Nominal-capacity status observed
	Nominal-capacity evidence reference P-ADVANCED
	Evidence class C
	P-ADVANCED

Named producer	Value
Tasnee	Process family polypropylene
	Published process route Unavailable
	Published standards Unavailable
	Observed nameplate capacity, t/y 720000
	Nominal-capacity status observed
	Nominal-capacity evidence reference P-TASNEE
	Evidence class C
P-TASNEE	

Observed adjacency signals

• Signal type	core_process
Description	large established Saudi polypropylene production
Evidence references	◦ P-ADVANCED◦ P-TASNEE
• Signal type	adjacent_output
Description	broad producer grade portfolios
Evidence references	◦ P-SABIC

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SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value	Unit
Simulated nameplate capacity	1170	kt

Source field	Value	Unit
	SYN-MINISTRY-PP-001::plant_line	
Availability factor	0.93 SYN-MINISTRY-PP-001::plant_line	ratio
Yield factor	0.97 SYN-MINISTRY-PP-001::plant_line	ratio
Qualification share	0.18 SYN-MINISTRY-PP-001::plant_line	ratio
Market allocation share	0.55 SYN-MINISTRY-PP-001::plant_line	ratio
Current utilisation	0.78 SYN-MINISTRY-PP-001::plant_line	ratio
Description	Unavailable Not established by the supplied evidence.	
Incremental capacity	Unavailable Not established by the supplied evidence.	kt
Upgrade schedule	Unavailable Not established by the supplied evidence.	months
Effective qualified capacity	104.49	kt
Qualified available supply	80	kt

Gap diagnosis and false-positive controls

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False or measurement gap

Source field	Value	Unit
Specification-adjusted gap	-24	kt

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SIMULATED — NOT MINISTRY EVIDENCE

Source reference	Status	Value
EX-01_HETEROGENEOUS_RESIDUAL	Not calculable	The commercial product cannot be separated from a highly heterogeneous residual code and no product-level evidence is available. SYN-MINISTRY-PP-001::hard exclusion inputs
EX-02_MARKET_BELOW_MES	Not calculable	The sustainable downside market is below minimum efficient scale and no credible export contract exists. SYN-MINISTRY-PP-001::hard exclusion inputs
EX-03_UNSATISFIABLE_HARD_GATE	Not calculable	A legal, safety, environmental, IP or customer-qualification hard gate cannot be satisfied. SYN-MINISTRY-PP-001::hard exclusion inputs
EX-04_IDLE_EQUIVALENT_CAPACITY	Satisfied	Domestic production meets the required specification with material idle capacity and no binding market failure is evidenced. SYN-MINISTRY-PP-001::hard exclusion inputs
EX-05_TRANSITORY_OR_MEASUREMENT	Not calculable	The apparent gap is dominated by re-export, one-off project demand, temporary price arbitrage or classification discontinuity. SYN-MINISTRY-PP-001::hard exclusion inputs
EX-06_REDUNDANCY_OR_CROWD_OUT	Not calculable	The intervention would create unacceptable redundant capacity or crowd out a more efficient incumbent. SYN-MINISTRY-PP-001::hard exclusion inputs

Test the apparent gap

Source reference	Status	Value
hard_exclusion	Satisfied	HARD_EXCLUSION_SATISFIED SYN-MINISTRY-PP-001::hard_exclusion_inputs
false_or_measurement_gap	Not calculable	FALSE_OR_MEASUREMENT_NOT_ESTABLISHED SYN-MINISTRY-PP-001::hard_exclusion_inputs
equivalent_idle_qualified_supply	Satisfied	EQUIVALENT_QUALIFIED_SUPPLY SYN-MINISTRY-PP-001::demand SYN-MINISTRY-PP-001::equivalence
uneconomic_at_efficient_scale	Not calculable	EFFICIENT_SCALE_ECONOMICS_UNAVAILABLE
structural_overcapacity	Not calculable	STRUCTURAL_OVERCAPACITY_NOT_ESTABLISHED SYN-MINISTRY-PP-001::hard_exclusion_inputs
generic_capacity_contradicted	Satisfied	GENERIC_CAPACITY_CONTRADICTED P-ADVANCED P-TASNEE P-WITS-390210

Capability and route alternatives

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Source field	Value	Unit
Known coverage K	1	ratio
Unknown weight U	0	ratio

Source field	Value	Unit
Uncertainty penalty (lambda)	0.5	ratio
Minimum known coverage	0.7	ratio
Published capability distance D*	0	ratio
Source field	State	Weight
Feedstock chemistry	0	0.15
Core process route	0	0.2
Equipment operating envelope	0	0.15
Finishing and specification control	0	0.1
Quality, laboratories and metrology	0	0.1
Certification and customer qualification	0	0.15
Capacity and timing	0	0.05
Utilities, environment and permitting	0	0.05
Skills and market integration	0	0.05
Sector-profile hard gates		
Polymer/additive compatibility	RESOLVED	
Conversion process	RESOLVED	
Tooling	RESOLVED	
Performance requirement	RESOLVED	
Application qualification	RESOLVED	
Unresolved hard gates	None recorded; no completed review is implied.	
Known hard-gate failures	None recorded; no completed review is implied.	

No intervention

Passes

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value 0

passes

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Information and market linkage

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Certification, testing and quality

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value	Not calculable
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Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Demand aggregation and offtake

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value	Not calculable
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Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Brownfield upgrade or expansion

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required.
- A lower-intervention route fully resolves the binding constraint.

Technology partnership or joint venture

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Greenfield capacity

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- The detected binding constraint is outside this route's intervention class.
- A lower-intervention route fully resolves the binding constraint.

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Public route comparison

No intervention

Passes

Source field	Value
Feasibility	Passes
Resolves the binding constraint	Passes
Additionality	Passes
Policy permissibility	Passes
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value 0

passes

- No action is the comparator and is selected when an evidenced rejection condition fully resolves the proposition.

Administrative or regulatory correction

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- Administrative, classification or regulatory barrier evidence is required.
- A lower-intervention route fully resolves the binding constraint.

Information and market linkage

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- A verified information, linkage or matching constraint is required.
- A lower-intervention route fully resolves the binding constraint.

Certification, testing and quality

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Incremental national value Not calculable

Not calculable

- A verified specification, certification, testing or qualification constraint is required.
- A lower-intervention route fully resolves the binding constraint.

Demand aggregation and offtake

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV Not calculable

Unsupported IRR Not calculable

Minimum effective support S* Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- A verified demand-fragmentation or offtake constraint is required.
- A lower-intervention route fully resolves the binding constraint.

Brownfield upgrade or expansion

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value	Not calculable
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Not calculable

- Verified incumbent adjacency makes brownfield the priority hypothesis; effective capacity and economics remain required.
- A lower-intervention route fully resolves the binding constraint.

Technology partnership or joint venture

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable

Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value	Not calculable
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Not calculable

- A verified technology constraint and major-line/JV adjacency are required.
- A lower-intervention route fully resolves the binding constraint.

Greenfield capacity

Fails

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No intervention

Incremental national value Not calculable

Not calculable

- Greenfield requires failed lower routes, D* above the configured band, demand at MES, and passing competition gates.
- A lower-intervention route fully resolves the binding constraint.

Shared enabling infrastructure

Not calculable

Source field	Value
Feasibility	Not calculable
Resolves the binding constraint	Not calculable
Additionality	Not calculable
Policy permissibility	Not calculable
Unsupported NPV	Not calculable
Unsupported IRR	Not calculable
Minimum effective support S*	Not calculable

Lower-route precedence: No lower route recorded

Incremental national value Not calculable

Not calculable

- GRAPH_REQUIRED: route 8 requires the governed dependency graph and a positive UnlockValue.

Counterfactual

Can the incumbent meet the specification without capital investment? Yes

Does downside demand support both incumbent and entrant? No

Does entry displace efficient domestic production? Yes

Is a technology partnership superior to expansion or greenfield? No

Missing capabilities • None recorded; no completed review is implied.

Brownfield versus greenfield comparison

Incremental capacity Unavailable

Upgrade schedule Unavailable

Unsupported NPV Unavailable

Minimum effective support S* 0

Greenfield alternative Unavailable

Economics and intervention

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SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value	Unit
Unsupported NPV	Not calculable Cannot be calculated from the supplied evidence.	SAR m
Unsupported IRR	Not calculable Cannot be calculated from the supplied evidence.	ratio
Minimum effective support S*	0	SAR m
Supported NPV	Not calculable Cannot be calculated from the supplied evidence.	SAR m
Supported IRR	Not calculable Cannot be calculated from the supplied evidence.	ratio
Hurdle rate	Not calculable Cannot be calculated from the supplied evidence.	ratio
Separate downside result	Not calculable Cannot be calculated from the supplied evidence.	SAR m

National-value comparison

Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Currency	SAR_m
Support required	0
Source explanation	No verified specification-adjusted gap; capacity support is non-additional.

[SYN-MINISTRY-PP-001::economics](#)

Competition and public policy

Class D simulation · SYN-MINISTRY-PP-001

SIMULATED — NOT MINISTRY EVIDENCE

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

Source field	Value	Unit
Capacity to downside demand ratio	Unavailable Not established by the supplied evidence.	ratio
Displacement cost	Unavailable Not established by the supplied evidence.	SAR m
Sunset date	Unavailable Not established by the supplied evidence.	
Triggers the default warning	Yes	
Non-additionality assessment		
Status	passes	
Recorded basis	route_record_additionality	
Concentration before and after	Not calculable	
Recorded finding	No new capacity is proposed; redundant support is avoided.	

Calculation and recommendation do not authorize public support.

Complete rule execution ledger

R0 · Identity gate

Full · Fires

HS revision H0 and HS6 390210 are frozen in the snapshot; specification/application remains separately gated.

Continue to screening while preserving the unresolved decision object.

R1-F · Persistent retained exposure — full

Disabled · Not evaluable

Full continuity is unavailable and gross flows are not retained imports.

Do not claim the full persistence rule.

Missing years	• 2022
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R1-D · Persistence signal — degraded

Degraded · Fires

3 positive observed years within the configured window; no interpolation used.

Generate INVESTIGATE only; confidence capped at C.

Positive observed years	• 2021 • 2023 • 2024
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Confidence ceiling	C
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R2 · Quantity-led expansion

Full · Does not fire

Value movement is not quantity-led under the configured test.

Separate structural volume growth from price movement.

First year	2023
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Final year	2024
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Observed period (years)	1
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Change in log value	0.0142
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Change in log quantity	-0.1793
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Change in log unit value	0.194
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Quantity contribution share	0.4804
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Quantity compound annual growth rate	-0.1642
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R3 · Supplier concentration

Disabled · Not evaluable

Value- and quantity-basis partner concentration are NOT_CALCULABLE.

No concentration inference.

Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Concentration threshold (HHI)	0.25
Largest-supplier threshold	0.5

Value

Recorded basis	value
Status	Not calculable
Observation year	2024
Trade-flow basis	gross
Source	none
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	Not calculable
Eligible partner count	Not calculable
Source explanation	No partner rows or source-attributed value concentration disclosure are available.

Quantity

Recorded basis	quantity
Status	Not calculable
Observation year	2024
Trade-flow basis	gross

Source	none
Concentration index (HHI)	Not calculable
Largest supplier share	Not calculable
Top-two supplier share	Not calculable
Valid observation coverage	Not calculable
Eligible partner count	Not calculable
Source explanation	No partner rows or source-attributed quantity concentration disclosure are available.

R4-F · Product-tier heterogeneity — full

Disabled · Not evaluable

Partner-month × tariff-line cells are unavailable.

Do not fit clusters or claim grades.

R4-D · Unit-value dispersion — degraded

Degraded · Fires

A source-attributed annual unit-value dispersion summary supports a descriptive product-mix signal; no cluster or grade conclusion.

Open specification research only; no grade conclusion.

Status	DISCLOSED
Calculation basis	disclosed_dispersion
Valid value coverage	Not calculable
Valid quantity coverage	Not calculable
Minimum valid-value coverage	0.7
Weighted first quartile (USD/t)	Not calculable
Weighted median (USD/t)	Not calculable
Weighted third quartile (USD/t)	Not calculable
Interquartile range (USD/t)	Not calculable

Bulk unit-value band (USD/t)	Not calculable
Potential outlier	Not calculable
Comparison observations	
• Trade flow	imports
Unit value (USD/t)	1649
• Trade flow	exports
Unit value (USD/t)	1097
Source evidence reference	P-WITS-390210
Source explanation	Gross-flow average unit values are a product-mix, grade, origin, or distribution research signal; they do not establish a specialty-grade gap.

R5 · Domestic supply plus continued imports

Degraded · Fires

Verified domestic capability coexists with material gross imports.

Test specification, qualification, capacity, price, application and allocation mismatch.

Retained imports (kt)	Not calculable
Net import exposure (kt)	Not calculable
Apparent consumption (kt)	Not calculable
Retained imports / apparent consumption	Not calculable
Calculation basis	Not calculable
Unavailable inputs	- domestic_origin_exports_kt • domestic_production_kt - reexports_kt • retained_imports_kt
Reasons for unavailability	
Retained imports (kt)	reexports_kt • retained_imports_kt
Net import exposure (kt)	domestic_origin_exports_kt • retained_imports_kt

Apparent consumption (kt)	- domestic_origin_exports_kt - domestic_production_kt - retained_imports_kt
Retained imports / apparent consumption	- apparent_consumption_kt - retained_imports_kt
Configured threshold	0.2
Source explanation	Domestic production quantity and retained-import flow are absent from the frozen public snapshot; gross imports cannot establish apparent consumption.

R6 · Capacity pressure

Disabled · Not evaluable

Effective utilisation and target-specification shortage are not public.

Obtain line-level availability, yield, qualification share and allocation.

R7 · Latent domestic capacity

Disabled · Not evaluable

Idle qualified capacity and specification equivalence are not established publicly.

Do not assume latent capacity.

R8 · Committed future demand

Disabled · Not evaluable

Awarded/financed demand at the target specification is unavailable.

Keep base, committed and announced demand separate when provided.

R9-S · Coarse incumbent adjacency screen

Full · Fires

A verified Saudi plant is in the same process family with additional capability signals.

Open the full line-level capability assessment; do not publish D* from screening alone.

Same process family	Yes
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Qualifying signal count	2
Known hard-gate failure	No

R10 · Strategic criticality

Disabled · Not evaluable

No responsible-authority criticality designation is present in the demo snapshot.

Keep commercial viability and strategic value separate.

Criticality status	Not calculable
Criticality evidence reference	Unavailable

R11 · Economic exclusion / generic-capacity warning

Full · Fires

Gross exports are 50.6× imports; generic new capacity cannot be justified from HS6 imports alone.

Reject generic support or move only a named specialty exception to investigation.

Export/import value ratio	50.6
Export/import ratio threshold	50
Calculated export/import value ratio	50.6013
Disclosed export/import value ratio	50.6
Disclosed ratio is consistent	Yes
Ratio basis	gross_trade_value
Ratio status	CALCULABLE
Ratio explanation	Unavailable
Established nominal capacity	
Established by evidence	Yes
Total observed nominal capacity (t/year)	1170000
Named producers	- Advanced Petrochemical - Tasnee

Evidence references	• P-ADVANCED• P-TASNEE
Source explanation	Unavailable

R12 · Evidence-value trigger

Degraded · Fires

5 named facts could change the route.

Prioritise the smallest evidence request with a plausible route effect; quantify EVSI when cost and value inputs exist.

Named missing facts	• Saudi tariff-line and invoice description • Importer/offtaker application and qualification requirements • Producer grade, application-equivalence and local-availability matrix • Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages • Delivered economics for the named specification/application exception
----------------------------	---

Evidence needed

- Saudi tariff-line and invoice description

Evidence-need identifier	identity/tariff-line
Blocked decision field	product_identity
Effect on route	Can resolve the decision object and change the admissible route.
Evidence references	◦ None recorded; no completed review is implied.
Numerical value of information	Not calculable

- Importer/offtaker application and qualification requirements

Evidence-need identifier	target specification/application
Blocked decision field	demand_at_required_specification
Effect on route	Can establish target demand and change state or route.
Evidence references	◦ None recorded; no completed review is implied.

Numerical value of information	Not calculable
• Producer grade, application-equivalence and local-availability matrix	
Evidence-need identifier	line-level production or producer-grade matrix
Blocked decision field	domestic_supply_or_capability
Effect on route	Can establish domestic equivalence and change the incumbent route.
Evidence references	◦ P-SABIC
Numerical value of information	Not calculable
• Effective availability, allocation, utilisation, yield, qualification share, backlog and planned outages	
Evidence-need identifier	capacity/availability/allocation
Blocked decision field	domestic_supply_or_capability
Effect on route	Can distinguish no action, brownfield expansion, and new capacity.
Evidence references	◦ P-ADVANCED◦ P-TASNEE
Numerical value of information	Not calculable
• Delivered economics for the named specification/application exception	
Evidence-need identifier	route economics
Blocked decision field	route_economics
Effect on route	Can determine whether the plausible route proceeds unsupported.
Evidence references	◦ P-WITS-390210
Numerical value of information	Not calculable

Simulated R6–R8 ledger

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

R5 · Retained import penetration

Disabled · Not evaluable

Synthetic retained-flow inputs are absent; R5 is not calculable.

Use only when production and retained-flow blocks are declared.

Retained imports (kt)	Not calculable
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Domestic production (kt)	Not calculable
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Gross import quantity	56
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Retained-import penetration	Not calculable
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Apparent consumption (kt)	Not calculable
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Minimum retained-import penetration	0.2
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R6 · Capacity pressure

Degraded · Does not fire

Synthetic R6 utilisation/shortage proxy does not meet both configured thresholds; sustained period is NOT_CALCULABLE.

Use as a DEGRADED capacity-pressure signal only; it cannot by itself support ADVANCE without a sustained period.

Effective utilisation	0.78
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Minimum effective utilisation	0.85
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Target-specification demand	56
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Effective qualified capacity	104.49
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Effective-capacity source field	effective_qualified_capacity_kt
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Shortage ratio	-0.4641
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Shortage denominator	effective_qualified_capacity_kt
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محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Minimum specification-matched shortage	0.1
Sustained period	Not calculable

R7 · Latent domestic capacity

Full · Does not fire

Synthetic utilisation/equivalence does not meet the configured latent-capacity test.

Test no-support, market linkage, procurement, or barrier removal only when equivalence and latent capacity are established.

Effective utilisation	0.78
Maximum effective utilisation	0.7
Specification equivalence	Yes
Qualified available supply	80

R8 · Committed future demand

Disabled · Not evaluable

Committed and announced layers are disclosed, but base demand, commitment probability, and minimum efficient scale govern R8 calculability.

Do not infer probability-adjusted demand addition or MES fill; keep committed and announced demand separate.

Base demand	Not calculable
Committed demand	0
Announced demand	0
Target-specification demand	56
Downside demand	51
Commitment probability	Not calculable
Probability-adjusted committed demand (kt)	Not calculable
Probability-adjusted demand addition	Not calculable

Minimum probability-adjusted demand addition	0.2
Minimum efficient scale (kt)	Not calculable
Minimum efficient scale coverage	Not calculable
Minimum scale-coverage threshold	0.25

Evidence passports and contradictions

4 public records; 12 synthetic records.

Contradiction register

Public evidence contradictions

No public contradictions recorded.

Synthetic evidence contradictions

No synthetic contradictions are recorded for this simulation.

P-WITS-390210

WITS/UN Comtrade Saudi trade rows for HS 390210

Source field	Value
Source	WITS/UN Comtrade
Class	B
Status	observed
Source period	2021/2023/2024
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Frozen annual world rows; export/import ratios and 2024 average unit values are disclosed calculations from the methodology worked case.
Supporting claims	• TRADE_VALUE• TRADE_QUANTITY• EXPORT_IMPORT_RATIO

Source field	Value
Contradiction register	Unavailable
Source reference	https://wits.worldbank.org/trade/comtrade/en/country/SAU/year/2024/tradeflow/Imports/partner/ALL/product/390210

P-SABIC

SABIC polypropylene portfolio

Source field	Value
Source	SABIC
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable
Supporting claims	DOMESTIC_PRODUCT_PORTFOLIO
Contradiction register	Unavailable
Source reference	https://www.sabic.com/en/products/polymers/polypropylene-pp

P-ADVANCED

Advanced Petrochemical company information

Source field	Value
Source	Advanced Petrochemical
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable

Source field	Value
Supporting claims	DOMESTIC_NAMEPLATE_CAPACITY
Contradiction register	Unavailable
Source reference	https://advancedpetrochem.com/about/

P-TASNEE

Tasnee petrochemicals page

Source field	Value
Source	Tasnee
Class	C
Status	observed
Source period	Unavailable
Retrieval date	2026-08-31
Reviewer status	unconfirmed_by_responsible_authority
Transformation	Unavailable
Supporting claims	DOMESTIC_NAMEPLATE_CAPACITY
Contradiction register	Unavailable
Source reference	https://www.tasnee.com/en/products/petrochemicals

SYN-MINISTRY-PP-001::capability_states

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Capability States

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: capability_states
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::class_if_confirmed

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Class If Confirmed

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: class_if_confirmed
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::counterfactual

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Counterfactual

Source field	Value
Source	DEMO_GENERATOR

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: counterfactual
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::decision_specific_hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision Specific Hard Gates

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: decision_specific_hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::demand

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Demand

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: demand
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::economics

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Economics

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable

Source field	Value
Supporting claims	Simulation branch input: economics
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::equivalence

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Equivalence

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: equivalence
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::evsi

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Evsi

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة
SIMULATED — NOT MINISTRY EVIDENCE

Source field	Value
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: evsi
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::hard_exclusion_inputs

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Exclusion Inputs

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_exclusion_inputs
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::hard_gates

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Hard Gates

Source field	Value
Source	DEMO_GENERATOR

Source field	Value
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: hard_gates
Contradiction register	Unavailable

SYN-MINISTRY-PP-001::plant_line

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Plant Line

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: plant_line
Contradiction register	Unavailable

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Target Specification

Source field	Value
Source	DEMO_GENERATOR
Class	D
Status	synthetic
Source period	Unavailable
Retrieval date	Unavailable
Reviewer status	Unavailable
Transformation	Unavailable
Supporting claims	Simulation branch input: target_specification
Contradiction register	Unavailable

Conditions, kill conditions and next facts

Class D simulation · SYN-MINISTRY-PP-001

محاكاة — ليست بيانات أو أدلة صادرة عن الوزارة

SIMULATED — NOT MINISTRY EVIDENCE

Decision conditions

- Reopen only for a named specialty-grade exception supported by buyer evidence.

Kill conditions

- Any proposed generic-capacity support is stopped while equivalent qualified capacity is available.

Next evidence actions

- None

Source field	Value	Unit
Approximate EVSI	-2	SAR m
Route-change probability	0.08	
Value difference (SAR m)	25	
Evidence acquisition cost (SAR m)	3	
Delay cost (SAR m)	1	
Approximate EVSI	-2	
Positive result	No	
Next fact to establish	Obtain named specialty-grade exception only if a buyer identifies one	

Decision history and expert overrides

None recorded; no completed review is implied.

Authority and reproducibility

Authority and versions

Source reference	Value
Methodology	docs/authority/Industrial_Opportunity_Resolution_Methodology_Final_KSA.docx 5717cbd42acc9947ce5e450013719275acb7ed1470847b21fb2cc547c8ac4ce9
Project	0.2.0
Thresholds	1.3.0
Sector profiles	1.1.0
Evidence policy	1.4.0
UI strings	1.7.0

Evidence boundary

Public decision uses public evidence only Yes

Synthetic branch isolation Yes

Scenario reconciliation

Status PASS

Scenario identifier SYN-MINISTRY-PP-001

Opportunity identifier SAU-H0-390210

Public snapshot identifier PUBLIC-SAU-H0-390210-2026-08-31

Recorded checks

- **Rule identifier** target_spec_demand_within_public_imports
Source Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula target_spec_demand_kt <= latest_public_imports_kt
Recorded result PASS
Blocking check Yes

Source inputs

Target-specification demand 56

Latest public observation year 2024

Latest public imports (kt) 56

Source explanation 56 kt <= 56 kt.

- **Rule identifier** line_nameplate_within_disclosed_public_capacity
Source Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula plant_line.nameplate_kt <= sum(non_null installed_capacity_tpy) / 1000
Recorded result PASS
Blocking check Yes

Source inputs	
Line nominal capacity (kt)	1170
Disclosed installed capacity (t/year)	◦ Unavailable◦ 450000◦ 720000
Publicly disclosed nominal capacity (kt)	1170
Expansion assumption applied	No
Source explanation	1170 kt <= 1170 kt.
• Rule identifier	capacity_factors_within_unit_interval
Source	Core 06 §10 item 3 lines 226-233; Core 06 §6.2 lines 148-156; Core 05 §8 lines 197-203
Recorded formula	for every capacity factor: $0 \leq \text{factor} \leq 1$
Recorded result	PASS
Blocking check	Yes
Source inputs	
Availability factor	0.93
Yield factor	0.97
Qualification share	0.18
Market allocation share	0.55
Current utilisation	0.78
Source explanation	All declared capacity factors are within [0,1].
• Rule identifier	qualified_availability_within_physical_output
Source	Core 06 §5.1 lines 92-101; Core 06 §10 item 3 lines 226-233
Recorded formula	$\text{qualified_available_kt} \leq \text{nameplate_kt} * \text{availability} * \text{yield}$
Recorded result	PASS
Blocking check	Yes
Source inputs	
Qualified available supply	80

Simulated nameplate capacity	1170
Availability factor	0.93
Yield factor	0.97
Physical output ceiling (kt)	1055.457
Source explanation	80 kt <= 1055.46 kt.
• Rule identifier	demand_layers_remain_separate
Source	Core 04 §2.6 lines 102-112
Recorded formula	observe downside_demand_kt <= target_spec_demand_kt and committed_demand_kt <= target_spec_demand_kt; no decision effect
Recorded result	INFORMATIONAL
Blocking check	No
Source inputs	
Target-specification demand	56
Downside demand	51
Committed demand	0
Base demand	Unavailable
Downside demand within target demand	Yes
Committed demand within target demand	Yes
Source explanation	Demand layers are reported separately; no ordering constraint is enforced.
• Rule identifier	tariff_line_allocation_sums_to_public_hs6_total
Source	Core 06 §5.1 lines 92-101; Core 05 §8 lines 182-195
Recorded formula	sum(tariff_line_allocation.lines.quantity_kt) == latest public imports_kt
Recorded result	Not applicable
Blocking check	Yes

Source inputs

Source explanation No tariff_line_allocation block is declared.

• **Rule identifier** buyer_allocation_within_public_imports

Source Core 06 §5.1 lines 92-101

Recorded formula sum(buyer_allocation.buyers.quantity_kt) <= latest public imports_kt

Recorded result Not applicable

Blocking check Yes

Source inputs

Source explanation No buyer_allocation block is declared.

• **Rule identifier** expansion_assumption_disclosed_and_bounded

Source Core 06 §5.1 lines 92-101; KL-38

Recorded formula planned_nameplate_kt >= disclosed public nameplate;
commissioning_year >= public as_of year; plant_line.nameplate_kt <= planned_nameplate_kt

Recorded result Not applicable

Blocking check Yes

Source inputs

Source explanation No expansion_assumption block is declared.

• **Rule identifier** retained_flows_reconcile_to_public_trade

Source Core 06 §5.1; methodology §3.3

Recorded formula retained_imports_kt <= imports_kt; retained + reexports == imports when reexports numeric; domestic_origin_exports_kt <= public exports_kt

Recorded result Not applicable

Blocking check Yes

Source inputs

Source explanation No production_and_retained_flows block is declared.

• Rule identifier	base_demand_and_commitment_probability_valid
Source	Core 06 §5.2; Core 07 §3 R8
Recorded formula	$0 \leq \text{commitment_probability} \leq 1$; $\text{base_demand_kt} > 0$; $\text{minimum_efficient_scale_kt} > 0$ when declared
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No base demand, commitment probability or MES block is declared.
• Rule identifier	shared_enabler_declaration_valid
Source	Core 06 scenario contract 2.1.0 and methodology §8.3
Recorded formula	explicit route valuation and complete Class-D declaration
Recorded result	Not applicable
Blocking check	Yes
Source inputs	
Source explanation	No shared-enabler declaration is present.

Latest imports (USD million)	92.3
Latest imports (kt)	56
Public decision unchanged after simulation	Yes
Scenario expected-result comparison	
Expected result	
State	REJECT
Route code	0
Actual result	
State	REJECT
Route code	0
Results match	Yes